

ProductivityBooster — Product Requirements Document (PRD)

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1. Problem Statement

Young adults — particularly students balancing academic and part-time work schedules, and mid-level engineers in meeting-heavy environments — spend significant time and mental energy manually managing their calendars. The core pain points are:

- **Schedule conflicts** are discovered too late and resolved manually, often by awkward last-minute cancellations
- **Meeting attendance** is inconsistent, with no lightweight system to track whether meetings were actually attended
- **Follow-ups** fall through the cracks because there is no prompt or system to capture action items immediately after a meeting ends
- **Calendar fragmentation** across Google, Outlook, and Apple Calendar means no single tool has the full picture
- **Declining meetings** requires manual drafting of messages, which is time-consuming and often skipped entirely

There is no tool today that combines intelligent conflict resolution, attendance tracking, follow-up management, and autonomous message drafting in a single, lightweight, privacy-respecting product.

2. Product Vision

ProductivityBooster is an AI-powered calendar assistant that helps users manage their time with minimal effort. It syncs across all major calendar platforms, detects and resolves conflicts intelligently, tracks meeting attendance, prompts follow-up capture, and eventually handles scheduling decisions autonomously — learning from each user's behavior over time.

North Star: A user should be able to trust ProductivityBooster to manage their schedule without manual intervention, confident that conflicts are resolved correctly, important meetings are never missed, and follow-ups are always captured.

3. Personas & Key User Journeys

PERSONA 1 — THE OVERWHELMED STUDENT

Name: Priya, 21

Context: Final-year undergraduate student working part-time at a café. Manages university lectures, group project meetings, work shifts, and personal commitments across Google Calendar and Apple Calendar. Frequently double-books herself and forgets follow-ups from study group sessions.

Goals: Never miss an important class or shift, spend less time untangling conflicts, get reminded about assignments or tasks discussed in meetings.

Frustrations: Notification overload, complex tools, feeling like the app is "too corporate" for her lifestyle.

Device: Primarily iPhone, occasionally MacBook.

Key User Journey — Priya:

1. Priya connects Google Calendar and Apple Calendar during onboarding (2 minutes).
2. She gets a conflict alert — her part-time work shift overlaps with a mandatory lecture.
3. The app surfaces the conflict, shows her past behavior ("You've prioritized lectures over work shifts 4/5 times"), and suggests keeping the lecture.
4. She approves. The app drafts a message to her manager: "Hi [Manager], I have a scheduling conflict and won't be able to make my shift on [date]. Can we discuss a reschedule?" — she approves it with one tap.
5. After her group project meeting ends, she gets a nudge: "Did you attend this meeting?" → Yes.
6. App asks: "Any follow-ups?" → AI suggests "Share meeting notes with group" based on the meeting title "Project Sync." She confirms and it creates a Notion task.
7. Weekly digest on Sunday shows her 3 pending follow-ups and upcoming conflicts for the week.

PERSONA 2 — THE MID-LEVEL ENGINEER

Name: Arjun, 32

Context: Software engineer at a mid-size tech company. Calendar is packed with standups, sprint planning, 1:1s, and ad-hoc client calls across Google Calendar and Outlook. Frequently has back-to-back meetings with no buffer. Spends 20+ minutes per week manually resolving conflicts and writing decline messages.

Goals: Spend less time in meetings that don't matter, never miss client-facing calls, have action items automatically captured and pushed to Jira.

Frustrations: Generic calendar apps with no intelligence, having to manually write "can't make it" emails, losing track of action items from meetings.

Device: MacBook primarily, Android phone.

Key User Journey — Arjun:

1. Arjun connects Google Calendar and Outlook during onboarding.
 2. He sets a conflict rule: "Client meetings always beat internal meetings."
 3. A sprint planning meeting is auto-declined because it conflicts with a client sync — the rule fires automatically, no input needed.
 4. For an ambiguous conflict (two internal meetings), the AI suggests keeping the one he's attended more consistently. He overrides and the override is remembered.
 5. After a client call, the app checks attendance and prompts: "Any follow-ups?" → He types "Send API docs to client." → App creates a Jira ticket automatically.
 6. Over time, the app learns Arjun always skips the Tuesday design review and begins auto-declining it with a pre-approved message template.
 7. Monthly behavioral report shows: 3 hours saved in manual scheduling, 94% follow-up completion rate.
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4. Business Success Metrics

These metrics define whether ProductivityBooster is viable as a business, measured at **6 months post-launch**:

Metric	Target
Monthly Active Users (MAU)	10,000+
Day-1 Retention	>60%
Day-30 Retention	>35%
Day-90 Retention	>20%
Free → Paid Conversion Rate	>8%
Average conflicts resolved per user/week	>3
AI decline messages sent (approved by user)	>2 per active user/week
Weekly digest open rate (email)	>45%
NPS Score	>40
Churn Rate (paid)	<5% monthly

5. Technical Success Metrics

These metrics define whether the product is working correctly under the hood:

Metric	Target
Calendar sync latency	<30 seconds from event creation to app detection
Conflict detection accuracy	>98% (no false positives on non-overlapping events)
AI conflict suggestion acceptance rate	>70% (measures quality of AI recommendations)
Follow-up prompt delivery rate	>99% (attendance nudge fires within 5 min of meeting end)
Notification delivery success rate	>99.5% (push + email)
AI decline message approval rate	>80% (users approve without editing)
Behavioral model accuracy	Conflict auto-resolution matches user preference >85% after 30 days
App load time (web)	<2 seconds on standard broadband
App load time (mobile)	<1.5 seconds on 4G
Uptime	>99.9%
Data deletion request completion	<24 hours

6. Feature Scope — MVP

IN SCOPE

Calendar Sync

- Connect Google Calendar, Microsoft Outlook, Apple Calendar via OAuth
- Real-time sync — detect new events, edits, and deletions within 30 seconds
- Unified view across all connected calendars

Conflict Detection & Resolution

- Detect overlapping events automatically
- Three-layer resolution: user-defined rules → AI suggestion (with past behavior) → manual override
- Learn from every override for future auto-resolution
- Long-term goal: fully autonomous resolution without user input

Attendance Tracking

- Push notification + email nudge 5 minutes after meeting end: "Did you attend?"
- Binary Yes/No response
- Attendance data feeds the behavioral learning engine

Follow-Up Management

- Post-attendance prompt: "Any follow-ups?"
- Free-text entry field
- AI-suggested follow-ups based on meeting title and description
- Push to Notion, Jira, or native calendar tasks (user selects integration)

AI Decline Message

- Draft message using user-defined excuse templates
- Suggest reschedule time based on free calendar slots (user must approve reschedule)
- Draft + one-tap approve flow; user can edit before sending
- Explicit one-time permission required before first send

Notifications

- Meeting reminder: 15 minutes before event
- Attendance nudge: 5 minutes after event end
- Weekly follow-up digest: user-configurable day/time
- Channels: mobile push notifications + email
- Frequency: user-configurable + AI-adaptive (reduces noise for consistent behaviors)

Behavioral Memory

- Stores: meeting priority patterns, recurring conflict resolutions, follow-up completion rates
- Used to improve AI suggestions over time
- User can view, edit, and delete all stored behavioral data

Onboarding

- Progressive — minimal setup (connect calendar, set timezone)
- Contextual feature discovery at first encounter of each feature
- Persona selection optional ("Student" / "Engineer") for pre-configured defaults

Platforms

- Web app (responsive, works on desktop and mobile browser)
- Mobile (iOS and Android — progressive web app for MVP, native app post-MVP)

7. Freemium Tier Breakdown

Feature	Free	Premium
Calendar connections	1	All 3 (Google + Outlook + Apple)
Meeting reminders	Yes	Yes
Basic conflict detection	Yes	Yes
AI conflict resolution	Yes	Yes
Behavioral memory (basic)	Yes	Yes (full history + detailed reports)
Attendance tracking	Yes	Yes
Follow-up free-text entry	Yes	Yes
AI-suggested follow-ups	Limited (3/month)	Unlimited
Task tool integrations	1 basic integration	Notion + Jira + native calendar
AI decline messages	5/month	Unlimited
Excuse templates	2 templates	Unlimited templates
Reschedule suggestions	Manual	AI-powered with calendar awareness
Weekly digest	Yes	Yes + monthly behavioral report
Data history	30 days	Unlimited
Notification customization	Basic	Full AI-adaptive control
Support	Community	Priority

Pricing Guidance (to validate with market research):

- Free tier: permanent
- Premium: ~\$8–12/month or ~\$80/year
- Student discount: 50% off premium (requires .edu email verification)

8. Notification System

TIMING

Trigger	Notification	Channel
15 min before meeting	"Your meeting [Title] starts in 15 minutes"	Push + Email
5 min after meeting ends	"Did you attend [Title]?"	Push
Follow-up not completed after 7 days	"You have [N] pending follow-ups"	Push + Email
Conflict detected	"You have a scheduling conflict on [Date]"	Push + Email
AI decline message ready	"Draft ready: decline message for [Meeting]"	Push

FREQUENCY CONTROLS

- **User-configurable:** Users can set quiet hours, disable specific notification types, and choose email digest frequency (daily/weekly)
- **AI-adaptive:** After 30 days, the system reduces reminders for meetings the user consistently attends and increases nudges for meetings they frequently miss or skip

DO NOT DISTURB

- User-defined quiet hours (e.g. 10pm–7am)
- Focus block detection — if a "Deep Work" or "Focus" block is on the calendar, suppress non-critical notifications during that window

9. Conflict Resolution System

THREE-LAYER RESOLUTION ENGINE

Layer 1 — Rule-Based (fires first)

- User defines rules during onboarding or settings (e.g. "Client meetings > internal meetings", "Lectures > work shifts")

- Rules are evaluated first — if a rule applies, conflict is resolved automatically
- User is notified of the auto-resolution but not required to act

Layer 2 — AI Suggestion (fires when no rule matches)

- AI analyzes behavioral history: which meeting types has the user attended more? Which have been skipped?
- Surfaces a ranked recommendation with reasoning: "Based on your history, you've attended standups 9/10 times vs. design reviews 3/10 times — keep standup?"
- User sees the suggestion with a one-tap approve or override option

Layer 3 — Manual Override (always available)

- User can override any AI suggestion
- Override is recorded and used to update the behavioral model
- Repeated overrides of the same pattern train the AI to make better suggestions

Autonomous Resolution (post-MVP north star)

- After sufficient behavioral data (suggested: 90 days), system offers to enable "Auto-resolve mode"
 - User approves enabling this mode explicitly
 - System resolves conflicts automatically, sends a notification summary of what was decided
 - User can always review and undo any auto-resolution within 1 hour
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10. AI Decline Message System

FLOW

1. Conflict is resolved (meeting A is dropped in favor of meeting B)
2. App asks: "Send a decline message to [Meeting A organizer]?" — user can skip
3. If yes: AI drafts message using the user's pre-saved excuse templates
4. App surfaces: reschedule suggestion based on next available shared free slot
5. User sees draft + reschedule time → one-tap approve OR edit before sending
6. User must have granted explicit send permission (one-time, revocable in settings)

MESSAGE TEMPLATES

- User defines 2 templates on free tier, unlimited on premium
- Templates support variables: [Organizer Name], [Meeting Title], [Date], [Suggested Reschedule]
- Example template: "Hi [Name], I have a conflicting commitment at [Date] and won't be able to join [Meeting Title]. Would [Suggested Reschedule] work for you instead?"
- Tone is not auto-adapted in MVP — user selects which template to use per message

RESCHEDULE LOGIC

- AI scans both user's calendar and (where available via calendar API) organizer's public availability
 - Suggests earliest mutually available slot within 5 business days
 - User must explicitly approve the reschedule time before it is included in the message
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11. Follow-Up Management

POST-MEETING FLOW

1. User confirms attendance ("Yes, I attended")
2. App asks: "Any follow-ups from [Meeting Title]?"
3. Free-text input field — user types freely
4. AI suggests 1-3 follow-up items based on meeting title/description (e.g. "Q2 Review" → suggests "Share summary", "Update forecast doc")
5. User selects AI suggestions and/or adds their own
6. Each follow-up is pushed to the user's selected task integration (Notion / Jira / native calendar task)

WEEKLY FOLLOW-UP DIGEST

- Every Sunday (default, user-configurable)
 - Lists all open follow-up items with meeting context
 - Shows completion rate trend: "You completed 7/10 follow-ups this week — up from 5/10 last week"
 - Delivered via push notification + email
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12. Behavioral Memory & Learning

WHAT IS STORED

Data Type	Purpose	Stored Where
Meeting attendance history	Improve conflict priority suggestions	Server (encrypted)
Conflict resolution choices	Train auto-resolve model	Server (encrypted)
Follow-up completion rate	Adapt nudge frequency	Server (encrypted)
Rule overrides	Refine rule suggestions	Server (encrypted)
Excuse template usage	Personalize message drafts	Server (encrypted)

USER CONTROL

- "My Data" screen shows all stored behavioral data in readable format
- User can delete any individual entry or all data at once
- Deletion is permanent and completed within 24 hours
- Opting out of behavioral memory disables AI suggestions but not core features

LEARNING TIMELINE

Days Active	Capability Unlocked
Day 1–7	Basic conflict suggestions (meeting type heuristics)
Day 8–30	Personalized suggestions based on user's own patterns
Day 31–90	High-confidence suggestions; override rate drops
Day 90+	Eligible for Auto-resolve mode (user opt-in)

13. Onboarding Experience

PROGRESSIVE ONBOARDING FLOW

Step 1 — Immediate (Day 0, 2 minutes)

- Connect at least one calendar (required)

- Set timezone
- Optional: select persona ("I'm a student" / "I'm an engineer") for pre-configured defaults
- Done — user enters the app immediately

Step 2 — First Conflict Encountered

- Contextual tooltip: "You have a scheduling conflict — here's how to resolve it"
- Walks user through conflict resolution UI once
- Offers to set first rule

Step 3 — First Meeting Ends

- Attendance nudge fires — brief explainer: "We'll check in after meetings to track attendance"
- Follow-up prompt fires — brief explainer: "Capture action items here — we'll remind you later"

Step 4 — First Follow-Up Pending (Day 7)

- Weekly digest arrives — explainer: "This is your weekly summary of open follow-ups"

Step 5 — First Decline Message Opportunity

- Explainer: "Want us to draft a decline message for you?"
- Walks through template setup (minimum 1 template required to enable feature)
- Requests explicit send permission

Never shown upfront: Behavioral reports, auto-resolve mode, advanced notification settings — these surface only when relevant.

14. Technical Considerations

CALENDAR INTEGRATIONS

- **Google Calendar:** Google Calendar API v3 — OAuth 2.0, push notifications via webhook (Google Calendar Watch)
- **Microsoft Outlook:** Microsoft Graph API — OAuth 2.0, real-time sync via subscription webhooks
- **Apple Calendar:** CalDAV protocol — polling-based sync (Apple does not support webhooks); sync interval: 30 seconds
- Token refresh must be handled silently — expired tokens should not log the user out

TASK TOOL INTEGRATIONS

- **Notion:** Notion API — OAuth, create database entries in user-selected database
- **Jira:** Jira REST API — OAuth 2.0 (3LO), create issues in user-selected project
- **Native Calendar Tasks:** Platform-specific (Google Tasks API, Microsoft To Do API, Apple Reminders via CalDAV)

AI & MACHINE LEARNING

- Conflict suggestion engine: rule-based classifier + user behavior weighting (can start with heuristics, graduate to ML model post-MVP)
- Follow-up suggestion: LLM prompt using meeting title + description as context (Claude API recommended)
- Decline message drafting: LLM prompt using user template + calendar context
- All AI calls must be async — never block the UI waiting for AI response

BACKEND ARCHITECTURE

- API-first backend (REST or GraphQL)
- Event-driven architecture for calendar sync and notification delivery
- Behavioral data stored in user-scoped encrypted database
- Message send actions require a separate permission scope — never bundled with calendar read permissions

PLATFORM

- **Web:** Responsive web app — React or Next.js recommended
- **Mobile:** Progressive Web App (PWA) for MVP — enables push notifications on iOS/Android without a native app
- **Native apps:** Post-MVP roadmap item

DATA STORAGE & SECURITY

- Behavioral data: server-side, AES-256 encrypted at rest
 - Calendar tokens: stored in secure vault (e.g. AWS Secrets Manager), never in the database
 - No calendar event content stored on servers beyond metadata (title, time, attendees) needed for features
 - All data transmission: TLS 1.3+
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15. UI/UX Style Preferences

DESIGN DIRECTION

Based on existing prototypes, the preferred aesthetic is the "**Bloom**" direction — warm, organic, approachable:

- **Color palette:** Warm cream background (#FDF6ED), sage green accents (#4A7C59), coral highlights (#E07A5F), ink text (#2C2416)
- **Typography:** Nunito (body, UI elements) + Lora (headings, accents) — approachable but professional
- **Tone:** Calm and supportive, not corporate or clinical

UI PRINCIPLES

- **Simple first:** Every screen should have one primary action. No feature discovery clutter.
- **Notification-minimal UI:** The app does not need to be open to work — most interactions happen via notifications
- **Progressive disclosure:** Advanced settings are buried — new users should never feel overwhelmed
- **Mobile-first layout:** Even the web app should feel comfortable on a phone screen
- **Accessible:** WCAG 2.1 AA compliance minimum — sufficient contrast, tap target sizes $\geq 44\text{px}$

KEY SCREENS

1. **Dashboard** — today's schedule, upcoming conflicts, pending follow-ups count
 2. **Conflict resolution card** — surface one conflict at a time, show AI suggestion with reasoning, approve/override
 3. **Post-meeting modal** — attended? + follow-up capture in one flow
 4. **Decline message preview** — draft text + reschedule suggestion + approve/edit/cancel
 5. **My Data** — behavioral history, stored patterns, data deletion controls
 6. **Settings** — notification preferences, calendar connections, task integrations, excuse templates
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16. Privacy & Compliance

DATA MINIMIZATION

- Only store metadata needed for features (event title, time, attendees, location)
- Do not store full calendar event descriptions on the server unless needed for AI follow-up suggestions (and only transiently, not persisted)

COMPLIANCE REQUIREMENTS

Regulation	Applicability	Key Requirements
GDPR	EU users	Right to access, right to erasure, data portability, consent management
CCPA	California users	Right to know, right to delete, opt-out of data sale (no data is sold)
FERPA	Student users	No sharing of education-related calendar data with third parties

USER CONTROLS

- Full data export (JSON) available on request
- Full data deletion within 24 hours — cascades to all behavioral memory, message history, and calendar tokens
- Granular consent: users can consent to behavioral memory separately from core calendar features
- Revocable send permission: user can revoke AI message-sending permission at any time in settings

THIRD-PARTY DATA SHARING

- No user data shared with third parties for any purpose
- AI inference (LLM calls for follow-up suggestions, message drafting) must be made via API calls that do not train on user data — use Anthropic API with data privacy terms confirmed
- Analytics (if any): anonymized aggregate data only, no individual user data

MESSAGE SENDING

- Explicit one-time permission required before the first message is sent on a user's behalf

- Permission is scoped: user must re-confirm if a new calendar account is connected
 - Every sent message is logged in the app with a "Messages Sent" history the user can view and audit
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17. Corner Cases

CALENDAR & SYNC

- **Event created during an existing event:** Conflict detected even if the new event overlaps only partially (e.g. last 15 minutes of meeting A overlaps first 15 minutes of meeting B)
- **Event deleted after conflict resolution:** If a meeting is deleted externally after the conflict was resolved, notify the user that the conflict is now moot
- **All-day events:** Do not trigger conflict detection or attendance nudges — treated as informational blocks
- **Recurring event exceptions:** If one instance of a recurring meeting is moved/deleted, only that instance is affected — not the whole series
- **Multi-timezone events:** Always display in user's local timezone; calculate conflicts in UTC to avoid DST edge cases
- **Calendar sync failure:** If a calendar fails to sync for >5 minutes, show a non-intrusive warning banner — do not silently fail

CONFLICT RESOLUTION

- **Three-way conflict (3 meetings at the same time):** Surface all three, resolve pairwise — highest priority vs. second, winner vs. third
- **Rule contradiction (two rules fire for the same conflict):** Surface to user with explanation — "Two rules apply here, your input is needed"
- **User changes their mind after auto-resolve:** Allow undo within 1 hour of auto-resolution; after that, treat as a new manual decision
- **No behavioral data yet (new user):** Fall back to manual priority prompt — never make a low-confidence AI suggestion appear confident

ATTENDANCE & FOLLOW-UPS

- **User doesn't respond to attendance nudge:** After 24 hours, mark as "unknown" — do not keep nudging

- **Meeting ran over time:** If the meeting's end time has passed but the user is still in the meeting (detected by no response to nudge), re-send nudge 15 minutes later once
- **Follow-up task integration fails:** If Notion/Jira API call fails, save the follow-up locally and retry silently — never lose a follow-up entry
- **User marks "Did not attend" for a mandatory meeting:** No judgment — simply do not generate follow-up prompt; flag for behavioral record

MESSAGING

- **Organizer's email not available in calendar event:** Cannot send decline message — inform user and offer to copy draft to clipboard instead
- **Message send fails (email bounce/API error):** Notify user immediately — do not silently fail; offer retry
- **User edits template to remove required variables:** Validate template before saving — warn if [Name] or [Date] variables are missing
- **Reschedule slot no longer available by the time user approves:** Re-run slot finder at time of approval — show updated suggestion if slot changed

PRIVACY & DATA

- **User requests data deletion while a message send is pending:** Cancel the pending send, then delete data
- **Minor users (under 18):** Require parental consent flow — flag during onboarding if student persona is selected and birth year indicates minor
- **Account sharing:** Behavioral memory is per-account — if two people use the same account, data is commingled; warn users that the app is designed for individual use

18. Out of Scope (Post-MVP)

The following features are explicitly deferred to post-MVP releases:

Feature	Reason Deferred
Native iOS/Android apps	PWA sufficient for MVP; native adds significant build time
Freelancer/Manager personas	Secondary personas; validate core with students + engineers first
Asana, Linear, Trello, Todoist integrations	Notion + Jira cover MVP use cases
SMS notifications	Push + email sufficient; SMS adds cost and compliance complexity
Meeting recording/transcript integration	High complexity; separate product surface
Team/shared calendar conflict management	Requires multi-user data model; out of MVP scope
Auto-resolve mode (fully autonomous)	Requires 90 days of behavioral data to be trustworthy
Tone adaptation per contact type	User templates cover MVP; adaptive tone is a premium AI feature
Calendar analytics dashboard	Behavioral reports sufficient for MVP
Enterprise SSO / admin controls	Enterprise tier is post-MVP

This PRD was developed through structured product discovery for the AI PM Edge Bootcamp, Week 1.

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